🛍️ Workshop Challenge: Mall Customer Segmentation with K-Means

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🎯 Objective

In this challenge, you will act as a **data scientist working for a shopping mall**.

Your mission is to use **K-Means Clustering** to segment customers into meaningful groups based on their purchasing behavior.

This challenge will help you:

- Apply unsupervised learning in a real-world scenario
- Choose the correct number of clusters
- Visualize clusters clearly
- Interpret results like a business analyst

The goal is not just to write code, it is to **understand what the clusters represent**.

📂 Dataset

Dataset Name: Mall Customer Segmentation Data
Dataset Link: [Mall Customer Segmentation Data]

📊 Dataset Description

The dataset contains the following columns:

- **CustomerID** → Unique identifier for each customer
- **Gender** → Gender of the customer (Male / Female)
- **Age** → Age of the customer
- **Annual Income (k$)** → Annual income in thousands of dollars

- **Spending Score (1–100)** → Score assigned by the mall based on customer behavior and spending habits

The **Spending Score** reflects how much a customer spends and how engaged they are with the mall.

⚠️ Important Instructions

For this challenge, you must use ONLY the following two features:

- **Annual Income (k$)**
- **Spending Score (1–100)**

Create your feature matrix like this:

```
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
```

🧠 Tasks

🔹 Task 1 : Load and Explore the Data

You must:

- Load the dataset
- Display the first rows
- Check dataset shape
- Check for missing values
- Create a scatter plot:

  Annual Income vs Spending Score

✏️ Questions to Answer:

1. Do you visually see possible groups in the scatter plot?
2. Approximately how many groups does your intuition suggest?

3. Why is visual exploration important before applying clustering?

🔹 Task 2 : Feature Selection

Create your feature matrix:

```
X = df[['Annual Income (k$)', 'Spending Score (1-100)']]
```

Remember:

In unsupervised learning, we only work with **X (features),** there is no target variable

✏️ Questions to Answer:

1. Why are we not using CustomerID in clustering?
2. Why is there no target variable in unsupervised learning?
3. Why do you think we did not use the entire dataset (Age, Gender, etc.) for this challenge?
4. In what situation would using more features be useful?

🔹 Task 3 : Use the Elbow Method

You must:

- Run K-Means for K = 1 to 10
- Store inertia values
- Plot the Elbow Curve
- Choose the best K

✏️ Questions to Answer:

1. What is inertia in your own words?
2. Which K did you choose and why?
3. What happens if K is too small?

4. What happens if K is too large?

🔹 Task 4 : Apply K-Means

Using your chosen K:

- Fit KMeans
- Get cluster labels
- Extract centroids

No interpretation yet, just compute and prepare for visualizatio.

🔹 Task 5 : Visualize Clusters

Create a clean visualization:

- X-axis → Annual Income
- Y-axis → Spending Score
- Points colored by cluster
- Centroids shown as large red "X"

Your plot must be:

- Clear
- Well labeled
- Easy to understand

✏️ Questions to Answer:

1. Do the clusters look well separated?
2. Are the centroids located near the center of their groups?
3. If some clusters overlap slightly, what could that mean?

🔹 Task 6 : Interpretation

Now think like a **business analyst**.

You must answer:

1. How many clusters did you find?
2. Describe each cluster.
3. Give each cluster a meaningful name.
4. If you were the mall manager, what marketing strategy would you apply to each cluster?

Example ideas (you may create your own):

- High income + high spending → “Premium Customers”
- High income + low spending → “Careful Rich”
- Low income + high spending → “Impulsive Buyers”
- Low income + low spending → “Budget Customers”

Be creative but realistic.

This is where real learning happens.

📦 Deliverables

You must submit:

✅ One Kaggle Notebook

The notebook must include:

- Clean and organized code
- Elbow plot
- Final cluster visualization
- Written answers to all questions (in Markdown cells)
- Clear interpretation of clusters

Everything must be inside the same notebook.

No separate files are required !

🧠 What You Should Understand After This Challenge

By completing this challenge, you should clearly understand:

- What unsupervised learning means
- How K-Means works step by step
- How to choose the number of clusters
- Why feature selection matters
- How to interpret clustering results in a business context

🚀 Good luck. *by Derriche Rania*